Deepa Garg

2401029 - T12

EXPERIMENT NO. 01

Aim

Version control operations using Git.

LO Mapped :L02

LO2:To perform version control operations using Git and apply collaboration workflows for software development.

Theory

Git is a distributed version control system that enables developers to track changes made to project files while maintaining a complete history of modifications. It allows multiple developers to work on the same project simultaneously without overwriting each other's work. Every change is stored as a commit, making it easy to review previous versions, compare changes, and restore earlier states whenever required.

GitHub is a cloud-based platform that hosts Git repositories and enables remote collaboration. Developers can push their local repositories to GitHub, clone repositories, create branches, merge changes, resolve conflicts, and synchronize their work with other contributors. Branching allows different features to be developed independently, while merging combines these features into the main project. Merge conflicts occur when multiple branches modify the same section of a file, requiring manual resolution before the merge can be completed.

In this experiment, a Git repository was initialized, connected to a GitHub repository, and used to demonstrate repository initialization, user configuration, committing changes, branching, merging, merge conflict resolution, rollback, revert, restoring files, and viewing project history.

Execution Steps

1. Create a Project Folder

mkdir GitLab1

cd GitLab1

Creates a new project directory and moves into it.

2. Initialize Git Repository

git init

Initializes a new Git repository.

3. Configure Username

git config --global user.name "Deepa Garg"

Sets the Git username for commits.

4. Configure Email

git config --global user.email "your-email@example.com"

Associates an email address with Git commits.

5. Verify Configuration

git config --list

Displays the configured Git settings.

6. Create a Project File

echo "Git Collaboration Project" > project.txt

Creates a text file inside the repository.

7. Check Repository Status

git status

Displays the current status of the repository.

8. Stage Files

git add .

Adds all files to the staging area.

9. Create Initial Commit

git commit -m "Initial commit"

Creates the first commit in the repository.

10. Rename Branch

git branch -M main

Renames the default branch to **main**.

11. Connect Remote Repository

git remote add origin https://github.com/deepa-garg/Git_lab.git

Links the local repository with GitHub.

12. Verify Remote Repository

git remote -v

Displays the configured remote repository.

13. Push Main Branch

git push -u origin main

Uploads the local repository to GitHub.

14. Create Login Feature Branch

git checkout -b feature-login

Creates and switches to the login feature branch.

15. Modify Project File

echo "Login Module" >> project.txt

Adds login feature information.

16. Stage Changes

git add .

Stages the modified file.

17. Commit Login Feature

git commit -m "Added login module"

Commits the login feature.

18. Push Login Branch

git push origin feature-login

Uploads the login branch to GitHub.

19. Switch to Main Branch

git checkout main

Returns to the main branch.

20. Create Dashboard Branch

git checkout -b feature-dashboard

Creates a new branch for the dashboard feature.

21. Modify Project File

echo "Dashboard Module" >> project.txt

Adds dashboard feature information.

22. Stage Changes

git add .

Stages the modified file.

23. Commit Dashboard Feature

git commit -m "Added dashboard module"

Commits the dashboard feature.

24. Push Dashboard Branch

git push origin feature-dashboard

Uploads the dashboard branch.

25. Switch to Main Branch

git checkout main

Returns to the main branch.

26. Merge Login Branch

git merge feature-login

Merges the login feature into the main branch.

27. Merge Dashboard Branch

git merge feature-dashboard

Attempts to merge the dashboard branch. If both branches modify the same part of a file, Git reports a merge conflict.

28. Resolve Merge Conflict

Edit **project.txt**, remove the conflict markers, and save the correct content.

Then execute:

git add project.txt

Marks the conflict as resolved.

29. Commit Merge Resolution

git commit -m "Resolved merge conflict"

Creates a commit after resolving the conflict.

30. Push Updated Main Branch

git push origin main

Uploads the merged project to GitHub.

31. View Commit History

git log --oneline --graph --all

Displays the complete commit history with branch visualization.

32. View Branches

git branch -a

Lists all local and remote branches.

33. Roll Back Last Commit

git reset --soft HEAD~1

Removes the latest commit while keeping the changes staged.

34. Revert a Commit

git revert <commit-id>

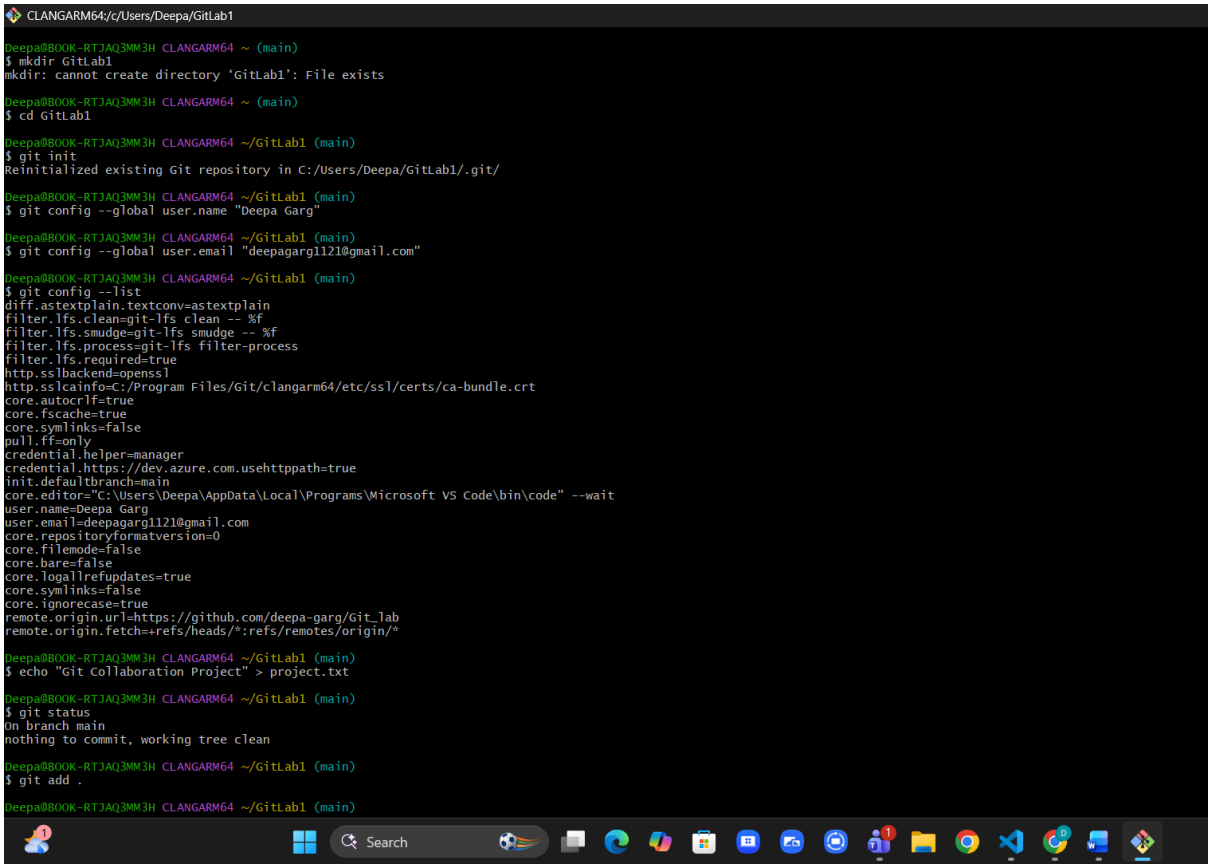
Creates a new commit that reverses the selected commit without changing the project history.

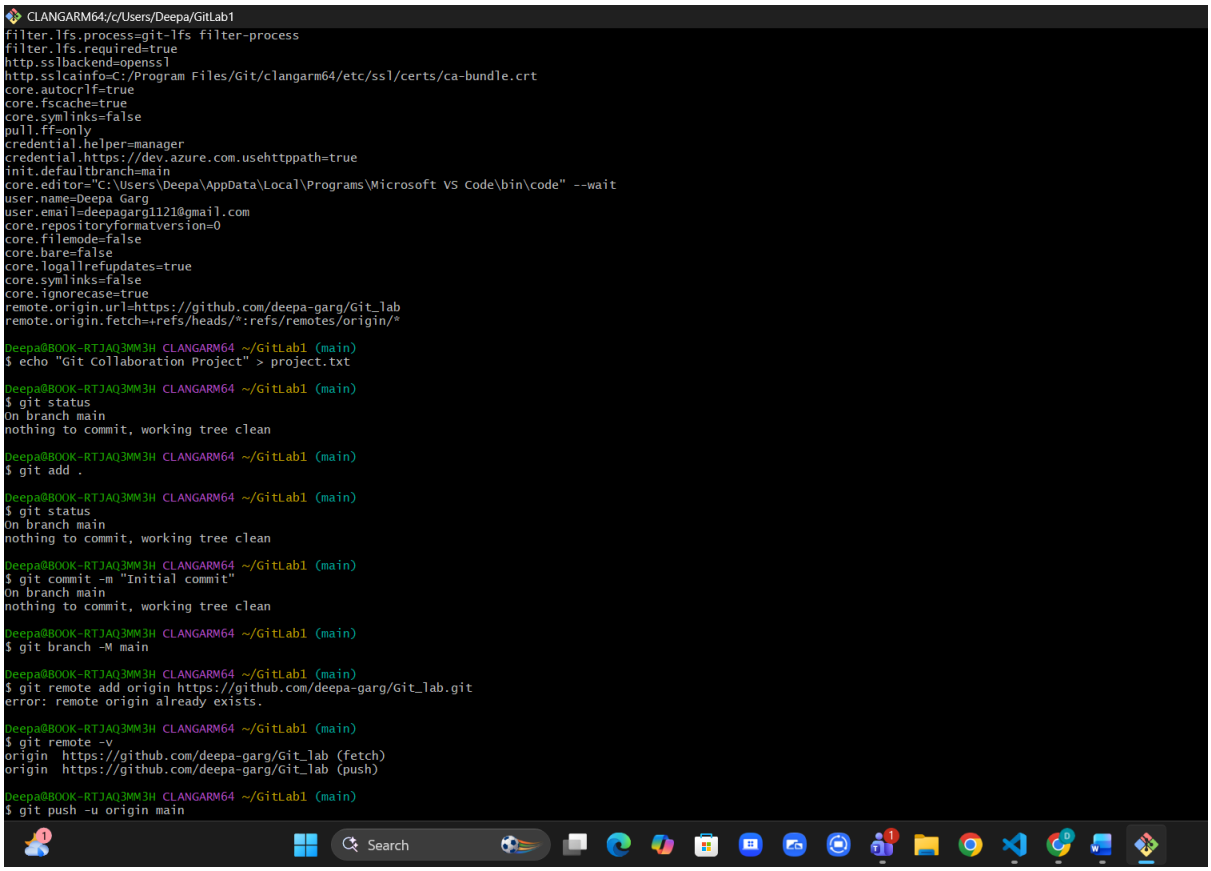
35. Restore Uncommitted Changes

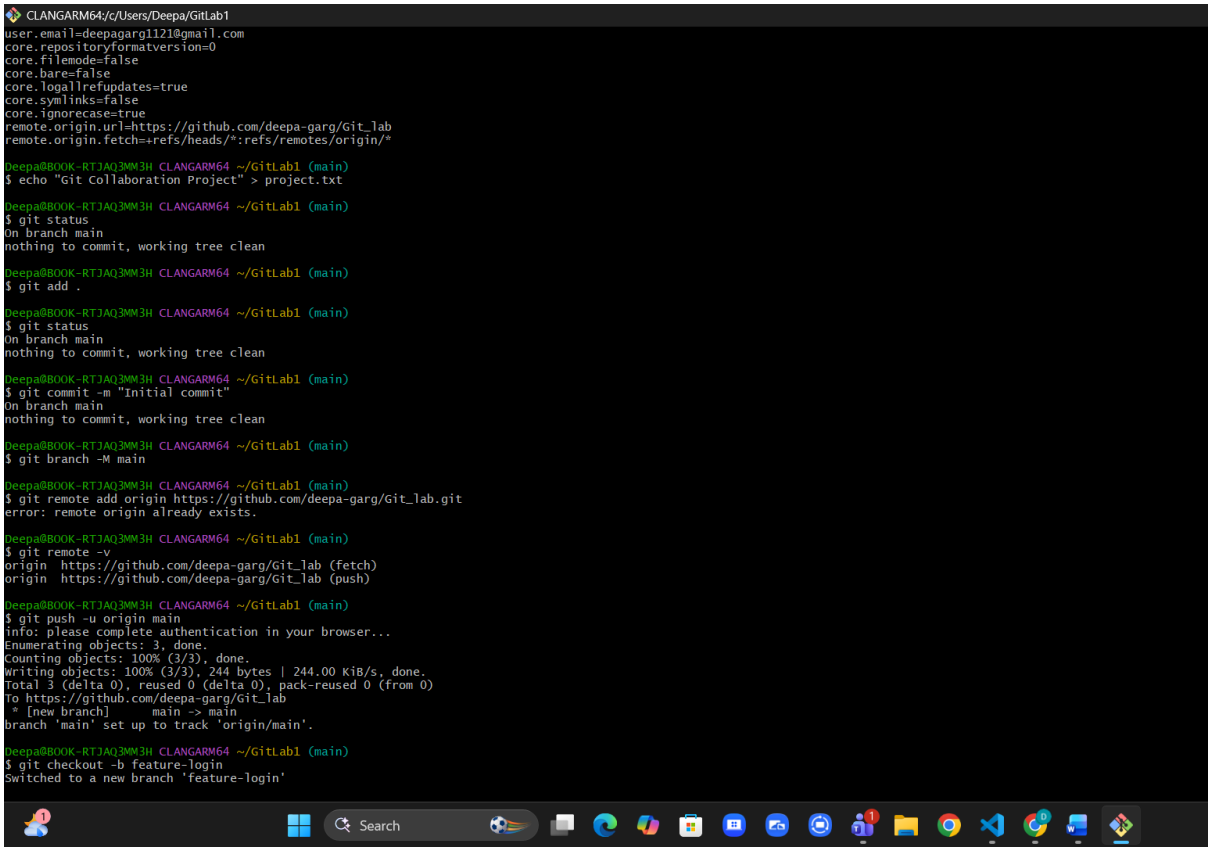
git restore .

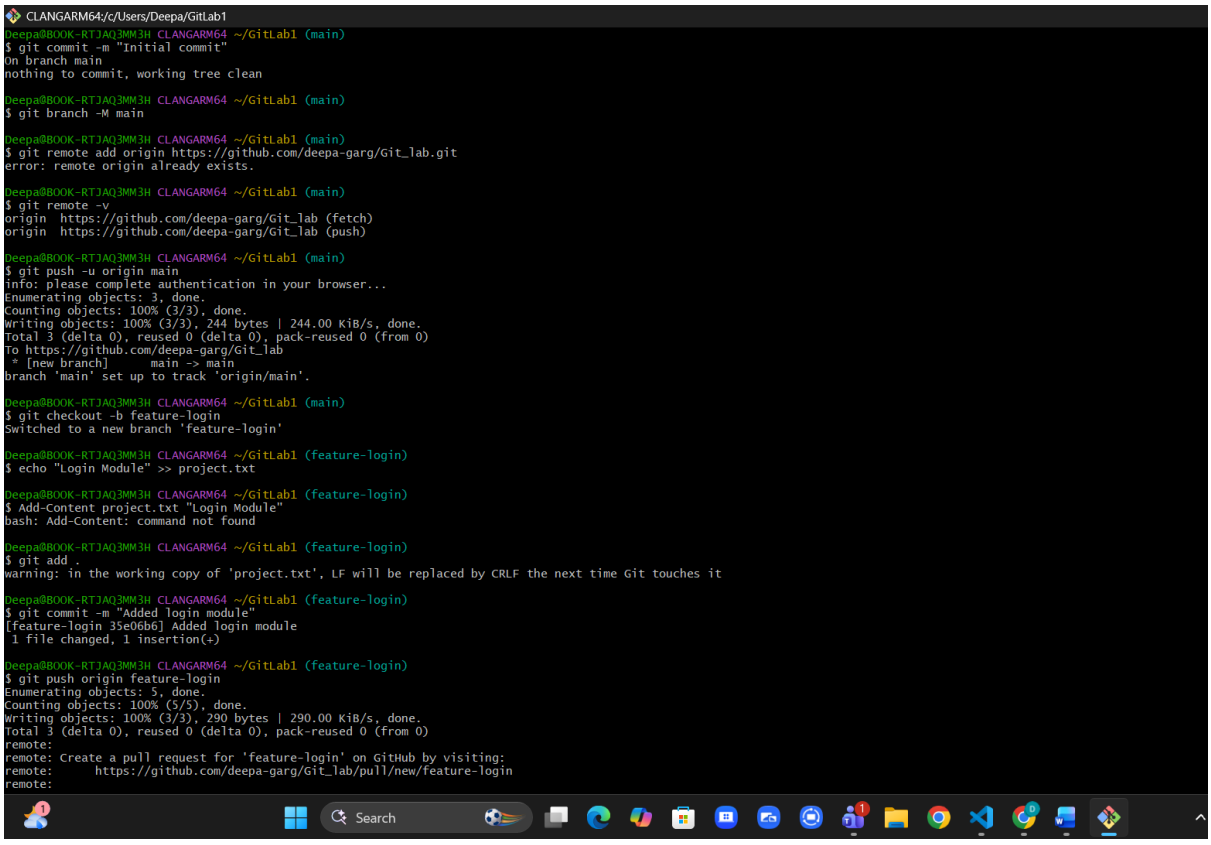
Discards all uncommitted modifications and restores the files to the last committed state.

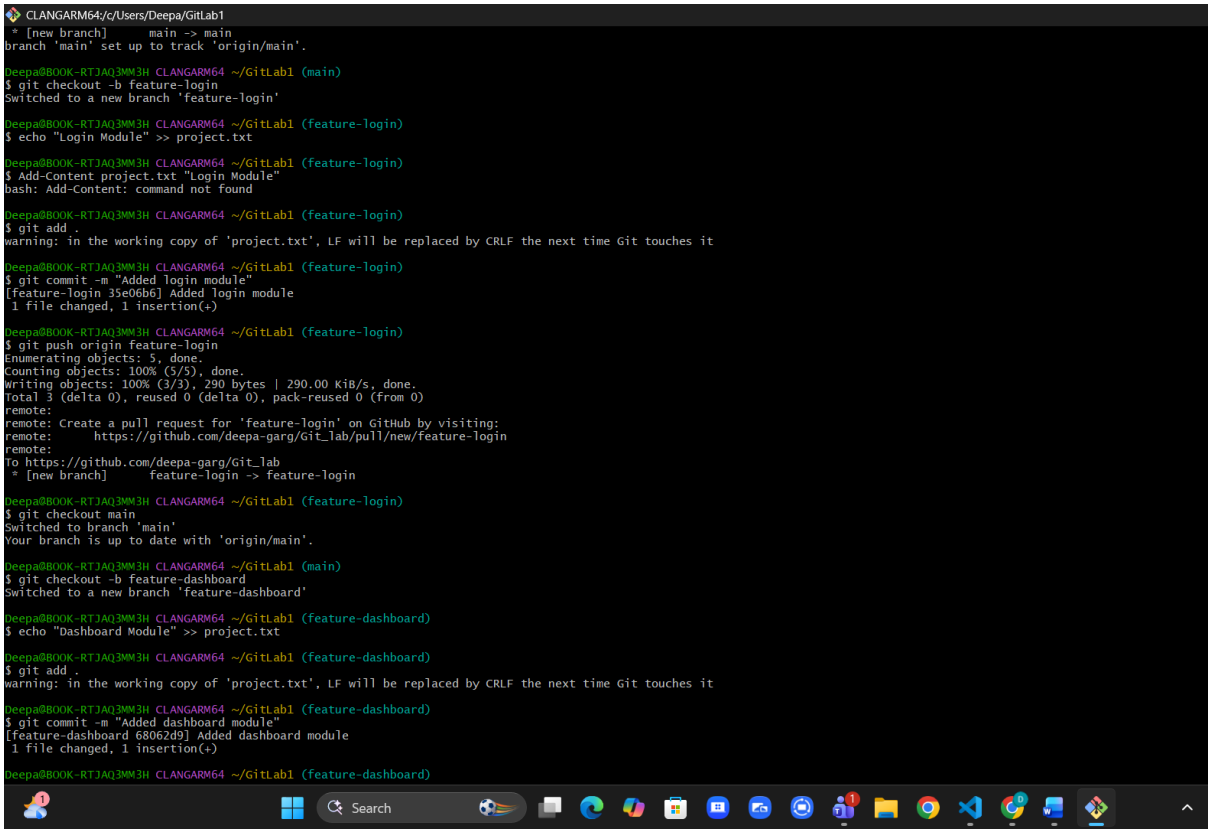
OUTPUT:

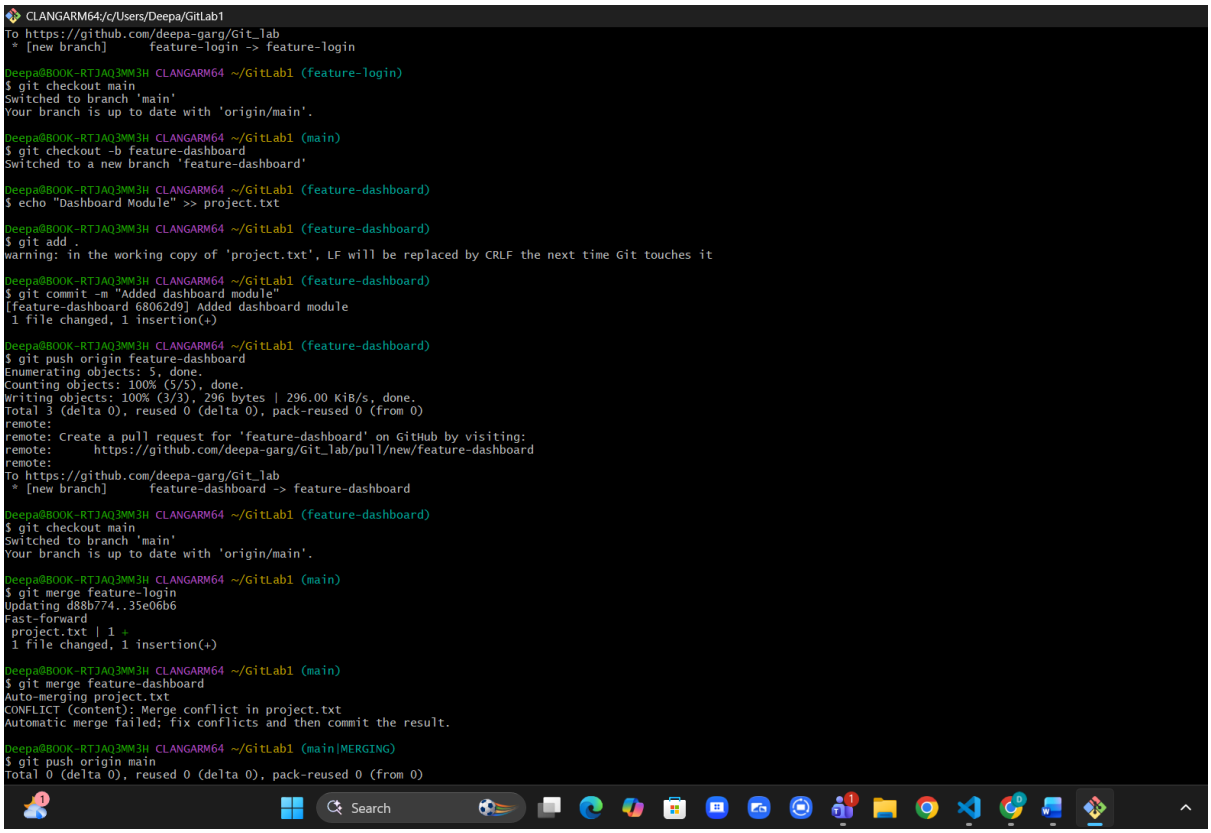

CLANGARM64:/c/Users/Deepa/GitLab1
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~ (main)
$ mkdir GitLab1
mkdir: cannot create directory 'GitLab1': File exists
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~ (main)
$ cd GitLab1
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git init
Reinitialized existing Git repository in C:/Users/Deepa/GitLab1/.git/
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git config --global user.name "Deepa Garg"
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git config --global user.email "deepagarg1121@gmail.com"
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/clangarm64/etc/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.ff=only
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=main
core.editor="C:\Users\Deepa\AppData\Local\Programs\Microsoft VS Code\bin\code" --wait
user.name=Deepa Garg
user.email=deepagarg1121@gmail.com
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
remote.origin.url=https://github.com/deepa-garg/Git_lab
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ echo "Git Collaboration Project" > project.txt
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git status
On branch main
nothing to commit, working tree clean
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git add .
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
Search


CLANGARM64:/c/Users/Deepa/GitLab1
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/clangarm64/etc/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.ff=only
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=main
core.editor="C:\Users\Deepa\AppData\Local\Programs\Microsoft VS Code\bin\code" --wait
user.name=Deepa Garg
user.email=deepagarg1121@gmail.com
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
remote.origin.url=https://github.com/deepa-garg/Git_lab
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ echo "Git Collaboration Project" > project.txt
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git status
On branch main
nothing to commit, working tree clean
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git add .
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git status
On branch main
nothing to commit, working tree clean
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git commit -m "Initial commit"
On branch main
nothing to commit, working tree clean
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git branch -M main
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git remote add origin https://github.com/deepa-garg/Git_lab.git
error: remote origin already exists.
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git remote -v
origin https://github.com/deepa-garg/Git_lab (fetch)
origin https://github.com/deepa-garg/Git_lab (push)
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git push -u origin main
Search


CLANGARM64:/c/Users/Deepa/GitLab1
user.email=deepagarg1121@gmail.com
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
remote.origin.url=https://github.com/deepa-garg/Git_lab
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ echo "Git Collaboration Project" > project.txt
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git status
On branch main
nothing to commit, working tree clean
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git add .
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git status
On branch main
nothing to commit, working tree clean
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git commit -m "Initial commit"
On branch main
nothing to commit, working tree clean
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git branch -M main
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git remote add origin https://github.com/deepa-garg/Git_lab.git
error: remote origin already exists.
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git remote -v
origin https://github.com/deepa-garg/Git_lab (fetch)
origin https://github.com/deepa-garg/Git_lab (push)
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 244 bytes | 244.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/deepa-garg/Git_lab
* [new branch] main -> main
branch 'main' set up to track 'origin/main'.
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git checkout -b feature-login
Switched to a new branch 'feature-login'
Search


CLANGARM64:/c/Users/Deepa/GitLab1
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git commit -m "Initial commit"
On branch main
nothing to commit, working tree clean
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git branch -M main
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git remote add origin https://github.com/deepa-garg/Git_lab.git
error: remote origin already exists.
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git remote -v
origin https://github.com/deepa-garg/Git_lab (fetch)
origin https://github.com/deepa-garg/Git_lab (push)
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 244 bytes | 244.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/deepa-garg/Git_lab
* [new branch] main -> main
branch 'main' set up to track 'origin/main'.
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git checkout -b feature-login
Switched to a new branch 'feature-login'
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-login)
$ echo "Login Module" >> project.txt
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-login)
$ Add-Content project.txt "Login Module"
bash: Add-Content: command not found
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-login)
$ git add .
warning: in the working copy of 'project.txt', LF will be replaced by CRLF the next time Git touches it
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-login)
$ git commit -m "Added login module"
[feature-login 35e06b6] Added login module
1 file changed, 1 insertion(+)
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-login)
$ git push origin feature-login
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Writing objects: 100% (3/3), 290 bytes | 290.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-login' on GitHub by visiting:
remote: https://github.com/deepa-garg/Git_lab/pull/new/feature-login
remote:
Search


CLANGARM64:/c/Users/Deepa/GitLab1
* [new branch] main -> main
branch 'main' set up to track 'origin/main'.
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git checkout -b feature-login
Switched to a new branch 'feature-login'
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-login)
$ echo "Login Module" >> project.txt
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-login)
$ Add-Content project.txt "Login Module"
bash: Add-Content: command not found
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-login)
$ git add .
warning: in the working copy of 'project.txt', LF will be replaced by CRLF the next time Git touches it
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-login)
$ git commit -m "Added login module"
[feature-login 35e06b6] Added login module
1 file changed, 1 insertion(+)
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-login)
$ git push origin feature-login
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Writing objects: 100% (3/3), 290 bytes | 290.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-login' on GitHub by visiting:
remote: https://github.com/deepa-garg/Git_lab/pull/new/feature-login
remote:
To https://github.com/deepa-garg/Git_lab
* [new branch] feature-login -> feature-login
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-login)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git checkout -b feature-dashboard
Switched to a new branch 'feature-dashboard'
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-dashboard)
$ echo "Dashboard Module" >> project.txt
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-dashboard)
$ git add .
warning: in the working copy of 'project.txt', LF will be replaced by CRLF the next time Git touches it
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-dashboard)
$ git commit -m "Added dashboard module"
[feature-dashboard 68062d9] Added dashboard module
1 file changed, 1 insertion(+)
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-dashboard)
Search


CLANGARM64:/c/Users/Deepa/GitLab1
To https://github.com/deepa-garg/Git_lab
* [new branch] feature-login -> feature-login
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-login)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git checkout -b feature-dashboard
Switched to a new branch 'feature-dashboard'
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-dashboard)
$ echo "Dashboard Module" >> project.txt
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-dashboard)
$ git add .
warning: in the working copy of 'project.txt', LF will be replaced by CRLF the next time Git touches it
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-dashboard)
$ git commit -m "Added dashboard module"
[feature-dashboard 68062d9] Added dashboard module
1 file changed, 1 insertion(+)
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-dashboard)
$ git push origin feature-dashboard
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Writing objects: 100% (3/3), 296 bytes | 296.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-dashboard' on GitHub by visiting:
remote: https://github.com/deepa-garg/Git_lab/pull/new/feature-dashboard
remote:
To https://github.com/deepa-garg/Git_lab
* [new branch] feature-dashboard -> feature-dashboard
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-dashboard)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git merge feature-login
Updating d88b774..35e06b6
Fast-forward
project.txt | 1 +
1 file changed, 1 insertion(+)
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git merge feature-dashboard
Auto-merging project.txt
CONFLICT (content): Merge conflict in project.txt
Automatic merge failed; fix conflicts and then commit the result.
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main|MERGING)
$ git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Search

```
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-dashboard)
$ git push origin feature-dashboard
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Writing objects: 100% (3/3), 296 bytes | 296.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-dashboard' on GitHub by visiting:
remote:      https://github.com/deepa-garg/Git_lab/pull/new/feature-dashboard
remote:
To https://github.com/deepa-garg/Git_lab
 * [new branch]      feature-dashboard -> feature-dashboard

Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (feature-dashboard)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git merge feature-login
Updating d88b774..35e06b6
Fast-forward
 project.txt | 1 +
 1 file changed, 1 insertion(+)

Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main)
$ git merge feature-dashboard
Auto-merging project.txt
CONFLICT (content): Merge conflict in project.txt
Automatic merge failed; fix conflicts and then commit the result.

Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main|MERGING)
$ git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/deepa-garg/Git_lab
   d88b774..35e06b6  main -> main

Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main|MERGING)
$ git log --oneline --graph --all
* 68062d9 (origin/feature-dashboard, feature-dashboard) Added dashboard module
| * 35e06b6 (HEAD -> main, origin/main, origin/feature-login, feature-login) Added login module
|/
* d88b774 Initial commit

Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main|MERGING)
$ git branch -a
  feature-dashboard
  feature-login
* main
  remotes/origin/feature-dashboard
  remotes/origin/feature-login
  remotes/origin/main

Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main|MERGING)
```



```
CLANGARM64:/c/Users/Deepa/GitLab1
Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main|MERGING)
$ git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/deepa-garg/Git_lab
   d88b774..35e06b6  main -> main

Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main|MERGING)
$ git log --oneline --graph --all
* 68062d9 (origin/feature-dashboard, feature-dashboard) Added dashboard module
| * 35e06b6 (HEAD -> main, origin/main, origin/feature-login, feature-login) Added login module
|/
* d88b774 Initial commit

Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main|MERGING)
$ git branch -a
  feature-dashboard
  feature-login
* main
  remotes/origin/feature-dashboard
  remotes/origin/feature-login
  remotes/origin/main

Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main|MERGING)
$ git reset --soft HEAD~1
fatal: Cannot do a soft reset in the middle of a merge.

Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main|MERGING)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

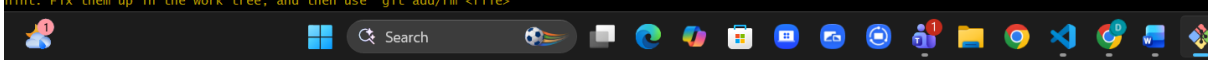
Unmerged paths:
  (use "git add <file>..." to mark resolution)
        both modified:   project.txt

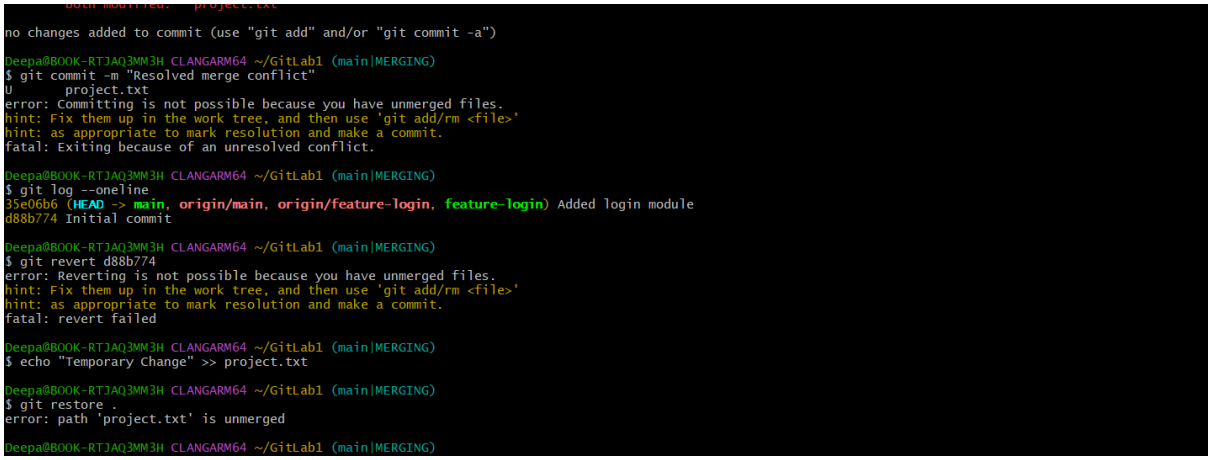
no changes added to commit (use "git add" and/or "git commit -a")

Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main|MERGING)
$ git commit -m "Resolved merge conflict"
U       project.txt
error: Committing is not possible because you have unmerged files.
hint: Fix them up in the work tree, and then use 'git add/rm <file>'
hint: as appropriate to mark resolution and make a commit.
fatal: Exiting because of an unresolved conflict.

Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main|MERGING)
$ git log --oneline
35e06b6 (HEAD -> main, origin/main, origin/feature-login, feature-login) Added login module
d88b774 Initial commit

Deepa@BOOK-RTJAQ3MM3H CLANGARM64 ~/GitLab1 (main|MERGING)
$ git revert d88b774
error: Reverting is not possible because you have unmerged files.
hint: Fix them up in the work tree, and then use 'git add/rm <file>'
```





Conclusion

A Git repository was successfully created and connected to GitHub. User credentials were configured, files were committed, feature branches were created and merged, merge conflicts were resolved, and version control operations such as rollback, revert, restore, and viewing commit history were performed successfully. The experiment demonstrated the effective use of Git and GitHub for version control and collaborative software development.